Nishanth D

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.85
GK Shetty Vivekananda Vidyalaya Ambattur Chennai TN	Class XII (CBSE)	2017	76.20%
GK Shetty Vivekananda Vidyalaya Ambattur Chennai TN	Class X (CBSE)	2014	10.00%

WORK EXPERIENCE

DISHA Ticketing Module -- IITH ERP | Solutions Architect

Led the module for a 3-person team (backend, frontend, QA); authored the core data models, transactional messaging, live config runtime, and access boundaries -- an event-driven microservice on FastAPI, MongoDB, Redis, and RabbitMQ

- Architected a **CAS-guarded state machine** as the single authorization gate for every ticket transition, using atomic find_one_and_update on expected_status so competing writes deterministically fail closed
- Built a **transactional outbox** (unique-indexed event IDs, exponential backoff, dead-letter parking) with **mandatory-publish detection**, preventing silent notification loss on unroutable messages or broker outages
- Designed a **Redis sorted-set delayed-job pipeline** with **per-job SET NX locking** across replicas and **MinIO retry/backoff**, guaranteeing zero duplicate purges and zero storage leaks
- Authored a single **scope-resolver** shared identically across REST, WebSocket, and Mongo queries -- closing an **IDOR vulnerability** with **404 existence-hiding** against ID probing
- Built a **TTL-bounded live config engine** materializing a **23-collection** config store into a lock-guarded snapshot with a **versioned audit trail**, propagating routing/RBAC changes in seconds with zero service restarts

PROJECTS

EvidenceRAG -- RAG Document Intelligence | Python, FastAPI, FAISS, SQLite, Ollama [GitHub]

- Built a **document-grounded RAG system** using **MiniLM embeddings**, FAISS semantic search, and **Qwen2.5 3B**, supporting PDF/TXT/Markdown ingestion and validated citations.
- Engineered a **persistent indexing pipeline** using SQLite/FAISS, supporting document deduplication, deletion, and restart recovery; added regression tests for index consistency and rebuild failures.
- Compared **chunking strategies** using Recall@k, MRR, and latency; achieved **93.75-100% Recall@5** on 24 development queries and diagnosed retrieval failures involving paraphrased questions.

End-to-End Android Malware Detection -- Static Analysis Pipeline [Github]

- Engineered a **high-throughput static-analysis pipeline** in **Python** using **Androguard** to extract manifest, **DEX bytecode**, and restricted-API features across **65K+ Android APKs**
- **Architected a memory-efficient sparse-matrix** featurization (**CSR**) with isolated train/test vocabularies and **schema fallbacks** to handle malformed binaries without pipeline halts
- Served predictions through an **ensemble classifier** (**XGBoost/RF**), achieving **F1 0.984 / ROC-AUC 0.997** and **<1% degradation under 10% label noise**

TransactiWar -- Secure Full-Stack Banking Application [Github]

- Built and hardened a **framework-free full-stack** banking app (**PHP/MySQL/Docker**) against the **OWASP Top 10**, deployed on **Azure**
- Owned the upload attack surface: blocked **decompression bombs**, **polyglot files**, and **TOCTOU**

races via **pre-decode dimension caps**, **GD re-encoding**, and **lock-guarded file replacement**
- Performed **offensive assessment** of rival teams' apps using **Nmap, Burp Suite, sqlmap, and Wireshark**

SKILLS

Languages: Python, C, C++, PHP, SQL, Bash

Tools & Platforms: Git, Linux, Docker, Wireshark, Nmap, Burp Suite, sqlmap

Backend: FastAPI, MongoDB, Redis, RabbitMQ, REST APIs, WebSockets

Networking: TCP/UDP Sockets, DNS, IPv4 & IPv6

Data & ML: pandas, NumPy, scikit-learn, XGBoost

SCHOLASTIC ACHIEVEMENTS

- Placed **2nd of 19 teams** in TransactiWar's live attack-defense war-game (Network Security course), sustaining **zero successful attacks** against the system over the week-long competition.

- Presented "Transferable, Auditable and Anonymous Ticketing Protocol" (ACM ASIA CCS '24) as coursework for Advanced Topics in Cryptology, scoring **second-highest in the class**.

RELEVANT COURSES

Advanced Topics in Cryptology | Cryptology | Network Security | Advanced Computer Networks | Software Verification | Advanced Data Structures and Algorithms | Foundations of Machine Learning

POSITIONS OF RESPONSIBILITY

Event Organizer -- Vibrance, VIT Chennai's Annual Cultural Fest